

The mass gap: current position of this programme

This is the synthesis of the notes written since the goal was set to a proof of the Yang–Mills existence and mass-gap conjecture. It states what has been proved here, what has been imported, what has been ruled out, and what remains, with constants explicit and with each claim labelled. The short version: the conjecture decomposes into six named statements, of which two are proved and one is imported; the **upper side of the problem, the clause $m < \infty$, is closed** as far as any variational method can take it, and reduced to the existence of the theory plus the nontriviality of one flowed correlator; the **lower side, the clause $m > 0$, is untouched**, and three natural routes to it have been closed by explicit computation rather than by difficulty. What remains is a controlled truncation at each scale, which is the constructive renormalization-group problem, and whose error is now explicit.

1. The decomposition

The obligations map turns “prove the mass gap” into six statements about the Kogut–Susskind Hamiltonian

$$H = \frac{\hbar c}{a} \left[\frac{g^2}{2} \sum_{\ell} (-\Delta_{\ell}) + \frac{2}{g^2} \sum_p (N - \text{Re tr } U_p) \right], \quad \Delta_{a,L} = \frac{\hbar c}{a} \delta(g; N_s, G).$$

	statement	status
T1	finite lattice: unique physical ground state, $\delta > 0$	proved here
T2	$\delta \geq \gamma(g^2/2)C_2$ for $g \geq g_0$, uniform in volume	proved here from an imported theorem
T2'	$\liminf_{N_s} \delta > 0$ for every $g > 0$	open; false for $U(1)$
T3	$\delta_{\infty}(g)/(a\Lambda_{\text{lat}}(g)) \rightarrow m/(\hbar c\Lambda) \in (0, \infty)$	open
T4	continuum infinite-volume theory with the axioms	finite-volume ultraviolet stability known
S	small volume: $\Delta = \delta_1 g^{2/3} \hbar c / L [1 + O(g^{2/3})]$	upper side proved, lower side conditional

The conjecture is T2' together with T3, given T4.

2. What is proved here

T1 (obligations map §2). For every a , $N_s \geq 2$, $g > 0$ and compact connected G : H has compact resolvent, a simple strictly positive ground state, a positive gap, and the ground state lies in the physical sector, which inherits the gap. Ingredients: Peter–Weyl for $-\Delta_{\ell}$, a bounded perturbation argument for compact resolvent, Feynman–Kac positivity, and invariance of the positive ground state under gauge transformations.

T2 in Hamiltonian form (T2 note). The electric term is a classical Hamiltonian in the Peter–Weyl partition with product ground state and unit local gap after normalization, and the magnetic term is a bounded translation-invariant perturbation of range $\{0, 1\}^3$ with

$$\beta = \frac{48N^2}{g^4(N^2 - 1)} \quad (SU(N)),$$

so Yarotsky’s gap-stability theorem gives, for $g \geq g_0 = (48N^2/[(N^2 - 1)\beta_*])^{1/4}$ and every lattice size,

$$\Delta_{a,L} \geq \gamma \frac{g^2}{2} \frac{N^2 - 1}{2N} \frac{\hbar c}{a},$$

with an infinite-volume ground state and exponential clustering.

A Lieb–Robinson bound (note). The unbounded part of H is a sum of commuting single-link Laplacians, so the interaction picture generated by it leaves a time-dependent bounded finite-range interaction of unchanged norm and support, and the standard theorem applies:

$$v \leq \frac{32 e N c}{g^2}, \quad \text{uniform in the volume,}$$

giving a strong-coupling correlation length $\xi \leq C' a N / (\gamma C_2 g^4)$, and showing that the lattice light cone opens faster than c whenever $g^2 < 32 e N$.

Upper bounds. The Feynman–Bijl inequality for any gauge-invariant $f(U)$ (note Proposition 1); the strong-coupling gap pinned to order $g^2 \hbar c / a$ on both sides (Corollary 3); the abelian gap bounded by the plaquette-sine structure factor (Theorem 5), which is how $U(1)$ becomes gapless; the proof that **no operator linear in the electric field with c-number coefficients is gauge invariant for a non-abelian group** (Proposition 6), so the abelian gap-closing channel is blocked by a string cost; and the averaged Polyakov-loop bound with its necessary condition on the transverse susceptibility (note).

The finiteness clause reduced. In the continuum,

$$m \leq \frac{M_{k+1}}{M_k} \quad \text{for every } k, \quad M_k = \int E^k d\rho(E),$$

with ρ the spectral measure of one flowed zero-momentum correlator, and the ratios decrease as k decreases (moment hierarchy). The best usable member,

$$m \leq \frac{\hbar C_0(0)}{\int_0^\infty C_0(\tau) d\tau},$$

needs only T4 and $C_0 \neq 0$. The free-field value of the weakest member is computed in closed form and checked twice (free-field note): $m \leq \frac{8}{3\sqrt{\pi}} \hbar c / \sqrt{t} \approx 4.26 \hbar c / \sqrt{8t}$, with the coupling cancelling.

Structure of the lower side. With $\mathcal{G} = \{g : \liminf_{N_s} \delta > 0\}$, T2' is $\mathcal{G} = (0, \infty)$, which holds if and only if $\mathcal{G}$ is open and closed; T2 gives nonemptiness, the Lieb–Robinson bound removes one obstacle to openness, and closedness is equivalent to the absence of a zero-temperature bulk phase transition, hence, on its second-order branch, to the uniqueness of the continuum limit (note).

3. What has been ruled out

Four statements, each by computation.

Perturbation around the free theory cannot give $m < \infty$ usefully. The flowed bound is $\approx 4.26 \hbar c / \sqrt{8t}$ in the perturbative regime, weaker than the expected gap by $1/(\sqrt{8t} \Lambda) \gg 1$; it reaches m only at $\sqrt{8t_*} \simeq 4.26 \hbar c / m$, where its coefficient stops being perturbative (free-field note §7).

Real-space blocking gains nothing. Yarotsky's proof is a polymer expansion, so the connected estimate a blocking step needs is available; applying his criterion to a lattice of blocks of M^3 sites gives

$$\beta_{\text{block}} = \frac{24 M^2 N}{g^2 \delta(g; M)}, \quad \frac{\beta_{\text{block}}}{\beta_{\text{direct}}} \simeq \frac{3 M^2}{8},$$

because the block boundary grows like M^2 while the block gap stays flat at strong coupling and falls like $1/M$ at small volume. Blocking is strictly worse than no blocking at every coupling (note).

Conjugation by the flow cannot help. The gradient flow is a diffeomorphism of the compact configuration space, so it is implemented by a unitary and conjugation preserves the

spectrum; the magnetic term becomes the flowed action and the electric term becomes nonlocal with range $\sqrt{8t}$, and in the free case the two effects cancel mode by mode (e^{-2tk^2} against e^{+2tk^2}). A locality-based criterion appears to improve by e^{-4tk^2} only because it drops the locality hypothesis, so the gain belongs to the truncation that restores it (note).

No upper bound can give $m > 0$. Finiteness of all negative moments is necessary for a gap and insufficient: $d\rho = e^{-\hbar c\kappa/E} dE$ has every negative moment finite with $\inf \text{supp } \rho = 0$ (moment hierarchy §5).

The small-volume corner is an ultraviolet problem. The Schur error of the reduction to the constant modes has a Berry part of relative order $g^2 \log(\Lambda L)$, absorbed by the running coupling, and a cubic part whose energy slope is $g^2(\Lambda L)^2$, so the bare-fiber reduction holds only for $g \ll 1/N_s$: a fixed-lattice theorem, with the renormalized regime requiring a dressed vacuum (Schur note).

4. Where the non-abelian structure has been isolated

Three statements distinguish the groups, and any proof of T2' must use one of them.

1. *The commutator potential.* In the zero-momentum sector the potential is $\frac{g^2}{2} \sum_{i < j} |\vec{x}_i \times \vec{x}_j|^2$, identically zero for an abelian group; where it is present the transverse zero-point energy $\hbar g |\vec{x}_i| / \sqrt{m}$ confines the valley and produces the gap $\delta_1 \hbar^{4/3} g^{2/3} m^{-2/3}$ (G07 §3, claim C133).
2. *Gauge invariance blocks the photon channel.* No operator linear in E with c-number coefficients commutes with Gauss's law for a non-abelian group, so the excitation through which the abelian gap closes must carry a Wilson line whose electric cost does not vanish at long wavelength (upper-bound note Proposition 6).
3. *The one-loop valley potential.* On the torus, $U(a) = \frac{2\hbar c}{\pi^2 L} \Phi(La)$ with $\Phi(b) = \sum'_m (1 - \cos m \cdot b) / |m|^4$, whose $k = 0$ term is exactly the four transverse oscillators of the matrix model; for compact $U(1)$ the potential vanishes identically and the holonomy is free (valley note §3b).

All three say that the abelian theory has flat directions that nothing lifts.

5. What remains, and of what kind

obligation	kind	nearest tool
T2': closedness of $\mathcal{G}$	the conjecture	a change of variables at each scale
openness of $\mathcal{G}$	technical	gap stability without frustration-freeness
T3	the conjecture, given T2'	asymptotic-freedom control of $\delta_\infty(g)$
T4	construction	Balaban-type renormalization group
S lower side, renormalized	construction	dressed nonzero-mode vacuum
(F1)-free finiteness	reduced	nontriviality of one flowed correlator

The pattern across the notes is uniform: **every statement that is uniform in the cutoff is a renormalization statement.** The strong-coupling theorem is uniform in the volume at fixed cutoff; the small-volume theorem is fixed-lattice; the flowed bounds are finite at fixed lattice and need the flow's renormalization in the continuum. The programme has pushed each

of these to the point where the renormalization step is the only thing left, and has shown that two ways of avoiding it, blocking and perturbation around the free theory, do not work.

6. Assessment

The route that survives is the one sketched at the outset and now made precise: close the small-volume corner at fixed lattice; recognize the blocked theory at each scale as a theory of the same form with a larger coupling; iterate until the coupling reaches the strong-coupling threshold, where T2 applies with a gap uniform in volume. The number of steps is finite and logarithmic in the scale ratio, of order thirty for $SU(2)$ at a bare coupling $g_{UV}^2 = 1/2$ (blocking note §5). The content of each step is the change of variables, and that is exactly what has no substitute: the notes show that projection alone loses, that the connected estimate does not rescue it, and that the free theory is too far away to expand around.

What this programme has added is a map with constants: six named statements, two proved, the finiteness clause reduced to a single correlator, two routes closed with explicit numbers, and three places where the non-abelian structure is isolated in a form that a proof could use. The remaining step is the one that has been open since the problem was posed, and the Jacobian note states its error: Gaussian in the truncation range over the flow radius, times $e^{2t\|G\|_\infty}$, so the step is controlled exactly where the flowed field strength satisfies $\|G\|_\infty \lesssim \ell^{-2}$. The missing ingredient is a pointwise bound on that field strength, where the flow supplies only an L^2 bound.